

Lien CARTAYA

PhD Candidate in Mathematics

liencartaya.github.io ORCID: 0009-0000-2463-978X liencartaya98@gmail.com Talca, Chile



Mathematics PhD candidate specializing in representation theory, invariant theory, and non-commutative algebra. Interested in reflection groups, rational Cherednik algebras, symplectic reflection algebras, quotient singularities, diagonal coinvariant rings, and computational algebra.

Professional Experience

2024 - Present	PhD Candidate in Mathematics , Universidad de Talca, Chile Working on zero-fiber rings associated with quaternionic reflection groups Using tools from representation theory, invariant theory, symplectic reflection algebras, and computational algebra Python Macaulay2 LaTeX
2022 - 2024	Master in Mathematics , Universidad de Talca, Chile Thesis: <i>Non-commutative deformations of quotient singularities</i> Studied computational aspects of invariant theory and symplectic reflection algebras Macaulay2 LaTeX
2020 - 2021	Junior Assistant Professor , Universidad de La Habana, Cuba Teaching and academic support in undergraduate mathematics Preparation of written and computational material for mathematics courses Python LaTeX
Sep. 2026 - Present	Mathematics Tutor , IDI Tutoring Academy Online mathematics tutoring in English
2024 - 2026	Social Media Assistant , Institute of Mathematics, Universidad de Talca, Chile Part-time academic communication and support for institutional outreach
2021	Data Scientist , Mandao, La Habana, Cuba Part-time data analysis work using spreadsheets, Python, and web analytics tools Python Excel Google Analytics

Publications

- 2025 **Zero fibers of quaternionic quotient singularities.** *Revista Matemática Iberoamericana* 41, no. 3, 1461–1487. Joint work with S. Griffeth. DOI: 10.4171/RMI/1528.

Manuscripts in Preparation

- Zero fiber rings for rank 2 quaternionic reflection groups.** In preparation.
- Wonderful models of subspace arrangements for division rings.** Joint work with S. Griffeth and D. Guevara. In preparation.

Education

2024 - 2028	PhD in Mathematics , Universidad de Talca, Chile. Advisor: Stephen Griffeth. PhD Candidate since January 2025.
2022 - 2024	Master in Mathematics , Universidad de Talca, Chile. Thesis: <i>Non-commutative deformations of quotient singularities</i> . Advisor: Stephen Griffeth.
2016 - 2020	Degree in Mathematics , Universidad de La Habana, Cuba. Thesis: <i>Study on high-dimensional algorithms for the reduction of STRESS in Multidimensional Scaling</i> . Advisor: Elina Miret Barroso.

Awards & Grants

2026	Selected young researcher , 13th Heidelberg Laureate Forum, Heidelberg, Germany.
2026	Operational expenses for research , ANID, Chile.
2024 - 2028	ANID Beca de Doctorado Nacional , Chile.

Selected Presentations

- Sep. 2026 **What zero remember?** Next Gen Session, 13th Heidelberg Laureate Forum, Heidelberg, Germany.

Aug. 2026	Zero fiber rings for rank 2 quaternionic reflection groups. XXV Latin American Congress of Algebra, Lima, Peru.
Jul. 2026	A Coxeter-type bound for zero-fiber rings of quaternionic quotient singularities. CIME School on Algebraic and geometric combinatorics: Coxeter group theory and representation theory, Cetraro, Italy.
Apr. 2026	On bounds for the degree of the zero fiber of rank-2 quaternionic reflection groups. XXXVIII Jornada de Matemática de la Zona Sur, Universidad del Bío-Bío, Concepción, Chile.
Aug. 2025	The diagonal coinvariant ring for quaternionic reflection groups. X Encuentro Nacional de Algebra – EMALCA Argentina, Córdoba, Argentina.
Apr. 2025	About the diagonal coinvariant ring for quaternionic reflection groups. XXXVII Jornada de Matemática de la Zona Sur, Universidad Católica del Maule, Talca, Chile.
Jul. 2024	Zero fiber of quotient singularities. XXIV Coloquio Latinoamericano de Algebra, Santiago de Chile, Chile.
Jul. 2021	Application of multidimensional scaling to SARS-CoV-2 patients treated with Interferon. Poster, I Festival Nacional de Matemática y Computación, La Habana, Cuba.
Mar. 2020	Study and implementation of algorithms for high-dimensional multidimensional scaling. IX Encuentro de Matemática Numérica y Optimización, Cuba–Mexico, La Habana, Cuba.

Organization of Seminars and Conferences

Jan. 2026	Member of the Organizing Committee of the 2026 Summer School for Undergraduates in Mathematics, Instituto de Matemáticas, Universidad de Talca, Chile.
-----------	--

Teaching Experience

Sep. 2026 -	Mathematics Tutor , IDI Tutoring Academy. Online mathematics tutoring in English.
Mar. 2026	Instructor , Preparatory Mathematics course for engineering degrees, Universidad de Talca, Chile. Duration: one week.
2025	Lecturer , Universidad de Talca. Precalculus and Calculus for Commercial Engineering.
2022 - 2025	Teaching Assistant , Universidad de Talca. Statistics and Probability, Calculus, Linear Algebra, and Ordinary Differential Equations.
2017 - 2022	Teaching Assistant , Universidad de La Habana. Calculus, Linear Algebra, and Real Analysis.
Jan. 2023	Teaching Assistant , Summer School on finite subgroups of SL_2 by S. Griffeth, Universidad de Talca, Chile.

Outreach

Jun. 2026	Instructor, Club de Matemáticas Yachay Tech. Minicourse <i>An introduction to Gröbner bases</i> . Notes and videos, Universidad Yachay Tech, Ecuador.
Jan. 2026	Instructor , short course <i>Invisible Patterns: Changing without changing</i> , Campamento Hannah, Instituto de Matemáticas, Universidad de Talca, Chile. Mathematics outreach program for female high school students.

Schools, Workshops, and Conferences Attended

Sep. 2026	IA en la práctica matemática, Pontificia Universidad Católica de Chile, Santiago, Chile.
Jul. 2026	Algebraic Combinatorics and Finite Groups IV, Cetraro, Italy.
Jun.–Jul. 2026	CIME School: Algebraic and geometric combinatorics: Coxeter group theory and representation theory, Cetraro, Italy.
Mar. 2023	CIMPA School <i>Crossroads of Geometry, Representation Theory and Higher Structures</i> , Puerto Madryn, Argentina.

Languages

Spanish (Native), English (B2), Italian (A1).

Interests

Camping, organization of events, scientific outreach.

Referees

Prof. Stephen Griffeth. Institute of Mathematics, Universidad de Talca, Chile. sgriffeth@utalca.cl
Prof. María Ofelia Ronco. Universidad de Talca, Chile. mronco@utalca.cl
Prof. Nicolás Libedinsky. Department of Mathematics, Universidad de Chile, Chile. nlibedinsky@gmail.com